

Impact of environmental regulations in high-income countries on exports of pollution-intensive industries in low- and middle-income countries

Loucas Di Pillo, Lucas Deweys

Empirical Project : International Trade

Cioppa Kelly, J.Tharakan

HEC Liège

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Contents

1	Data & Sources	3
1.1	Research Question	3
1.2	Database Used	3
1.3	Variable Construction	4
1.4	Descriptive statistics	5
2	Empirical Motivation & Expected Effects	7
2.1	EPS in importing HIC	7
2.2	GDP of exporting LMIC	7
2.3	Distance	8
2.4	Inflation in LMIC	9
3	Econometric Model & Identification Strategy	10
3.1	Model Specification	10
3.2	Estimation Method	10
4	Results	11
4.1	Baseline model	11
4.2	Refined Analysis and Comparison	12
5	Limitations	13
6	Conclusion	14

1 Data & Sources

1.1 Research Question

The Pollution Haven Hypothesis (Wikipedia) states that high environmental standards in rich countries will increase the cost of production in industries that produce a lot of pollution, leading to the outsourcing of such productions to countries that have less stricter environmental standards. Consequently, low and middle-income countries will focus their export activities on the production of goods that produce a lot of pollution. The paper will investigate whether low and middle-income countries increase their export products that produce a significant amount of pollution when rich countries implement stringent environmental standards.

1.2 Database Used

To answer our research question, we combined several datasets providing information on bilateral trade flows, environmental regulations, and macroeconomic characteristics:

- Source: United Nations Comtrade Database. (Cometrade)
Variable extracted : Annual exports of pollution-intensives goods from low- and middle-income countries to high-income countries (value in USD).
Use in the model: Our dependent variable.
Details: We selected only HS codes classified as pollution-intensive in the scientific literature (1) (metals, chemicals, plastics, etc.).
- Source: OECD Environmental Policy Stringency database (9).
Variable extracted: Environmental Policy Stringency (EPS) index for high-income importing countries.
Use in the model: Main independent variable.
Details: Index ranges from 0 (not stringent) to 6 (very stringent), available annually.
Data was available only from 2010 to 2018.
- Source: World Development Indicators (WDI).
Variables extracted: GDP (current USD) for exporting LMIC (5); Inflation (CPI-based) for exporting LMIC (Group).
Use in the model: Control variables capturing market size and macroeconomic stability.
- CEPII GeoDist. (CEPII)
Variable extracted: Bilateral geographic distance (in km) between exporter and

importer.

Use in the model: Time-invariant control for trade costs.

We chose the United Nations Comtrade Database as the source of the data for bilateral trade, since it is considered the norm and standard source in empirical international trade studies. The OECD EPS dataset was used because it covers an important domain with regard to measuring environmental policy stringency. Finally, macroeconomic indicators include GDP and inflation, were taken from the World Bank WDI database, commonly used in empirical papers for its consistency and international comparability, while geographic distance was obtained from the CEPII GeoDist dataset, the benchmark source in gravity-model applications.

1.3 Variable Construction

To construct the variables used in the empirical analysis, we merge several datasets to build a panel of bilateral trade flows between low- and middle-income exporting countries (LMICs) and high-income importing countries (HICs). Since the Environmental Policy Stringency (EPS) index is available only for the period 2010-2018, the final sample is restricted accordingly.

As Teaching Assistant recommended us to ensure manageable data size and improve data quality, the sample is further restricted to LMIC exporters whose cumulative exports of pollution-intensive goods exceeded USD 10 billion over the 2010-2018 period. This restriction excludes countries with very small trade flows, which are more likely to introduce noise and complicate data management without contributing meaningful variation.

- **Dependent variable :** The dependent variable is the annual value of exports of pollution-intensive goods (in USD) from LMIC exporters to HIC importers. Pollution-intensive products are identified using the following Harmonized System (HS) chapters: HS 25 (Salt, sulphur, earths and stone), HS 26 (Ores, slag and ash), HS 27 (Mineral fuels, oils, and bituminous substances), HS 28 (Inorganic chemicals), HS 29 (Organic chemicals), HS 47 (Pulp of wood or other fibrous material), HS 48 (Paper and paperboard), HS 49 (Printed books and printed products), HS 72 (Iron and steel), and HS 73 (Articles of iron or steel). Export values are aggregated at the bilateral country-pair-year level and transformed using the natural logarithm to reduce skewness and allow coefficient estimates to be interpreted in percentage terms.
- **Independent variable :** For each importing high-income country is assigned its annual EPS index.

- Control variables: The set of control variables includes the exporting LMIC's gross domestic product (GDP) in current USD, which captures economic size and production capacity, and is expressed in logarithmic form for scale comparability. We also include the annual consumer price inflation rate of the exporting LMIC to control for macroeconomic stability and domestic cost conditions. In addition, bilateral geographic distance between exporting and importing countries is obtained from CEPII and included as a time-invariant proxy for trade costs.

Both export values and GDP exhibit strong right-skewness and vary substantially across countries. Following standard practice in empirical trade and macroeconomic research, these variables are log-transformed to mitigate the influence of extreme observations and to obtain more symmetric distributions (see (6)).

1.4 Descriptive statistics

Table 1 reports descriptive statistics for all variables used in the analysis.

The variable PrimaryValue (Exports) has a mean of 3.97×10^8 (approximately USD 397 million), while the median (p50) is substantially lower at 1.66×10^7 (USD 16.6 million). The large gap between the mean and the median indicates that export flows are highly right-skewed. This supports the use of a logarithmic transformation of exports in the empirical model, as it reduces skewness and limits the influence of extreme trade flows.

The mean value of Environmental Policy Stringency (EPS) is 3.07, with a standard deviation of 0.82. Given the 0–6 scale of the OECD EPS index, this average reflects a moderate level of environmental regulation among high-income importing countries. The observed range (approximately 1 to 5) and the standard deviation suggest sufficient variation in policy stringency to allow for meaningful econometric identification.

Similar to exports, GDP exhibits substantial right-skewness, with a mean of 3.19×10^{12} and a median of 5.58×10^{11} . Consequently, the logarithmic transformation of GDP is standard in gravity models and is appropriate in this context to prevent very large economies from disproportionately influencing the estimation results.

Geographic distance (DIST) has an average value of 7,990 kilometers, which is relatively large. This is consistent with the structure of the sample, which includes trade flows between high-income countries and low- or middle-income countries that are often geographically distant (for example, Europe and Southeast Asia, or North America and Africa).

Regarding missing data, most variables contain 4,997 observations, while inflation is available for 4,713 observations. Inflation data are missing for Argentina during the sam-

ple period. Argentina is nevertheless retained in the sample due to its importance as an exporter of pollution-intensive goods, and the resulting reduction in observations is limited.

Variable	N	Mean	p50	Range	SD	Variance
primaryValue	4997	3.97e+08	1.66e+07	2.15e+10	1.30e+09	1.69e+18
EPS	4997	3.071471	3.083333	4	.8209329	.6739309
gdp	4997	3.19e+12	5.58e+11	1.41e+13	4.67e+12	2.18e+25
DIST	4997	7990.777	8539.15	18441.71	3655.585	1.34e+07
inflation	4713	5.230753	3.597823	37.50346	5.442444	29.6202

Figure 1: Descriptive statistics (Stata output).

Figure 3 presents the distribution of the logarithm of the dependent variable, while 2 displays the distribution of the Environmental Policy Stringency (EPS) index.

The distribution of the dependent variable, $\log(\text{PrimaryValue})$, indicates that the logarithmic transformation effectively mitigates the strong right-skewness observed in the raw trade data. Although the transformed variable remains slightly left-skewed, with a long tail corresponding to lower-value trade flows, the overall distribution closely approximates a normal distribution, centered between 10 and 20. This transformation is important for stabilizing variance and increasing the likelihood that the OLS residuals satisfy the normality assumption required for valid statistical inference.

In contrast, the Environmental Policy Stringency (EPS) index exhibits a multimodal distribution, with distinct peaks around 3.0, 3.7, and 4.1. This pattern suggests that high-income countries in the sample tend to adopt environmental regulations in discrete steps rather than along a continuous spectrum.

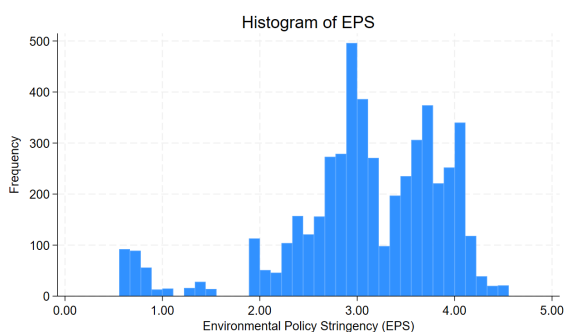


Figure 2: Distribution of EPS.

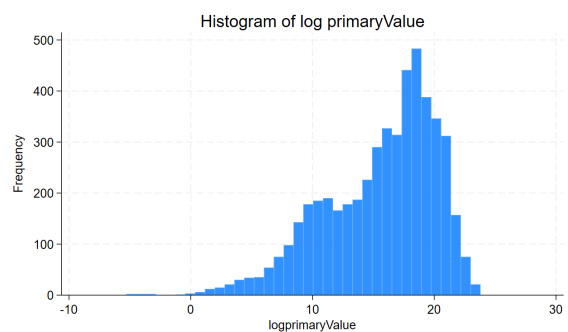


Figure 3: Distribution of $\log(\text{Exports})$.

2 Empirical Motivation & Expected Effects

2.1 EPS in importing HIC

For instance, environment regulation in high-income countries can affect foreign suppliers' competitiveness, particularly in pollution-intensive sectors. Tightening home-country policies raises the cost of production for domestic firms, and hence, can shift demand to foreign suppliers, including those from LMIC exporters (Kellenberg, 2009 (7)). This mechanism often relates to the pollution-haven hypothesis.

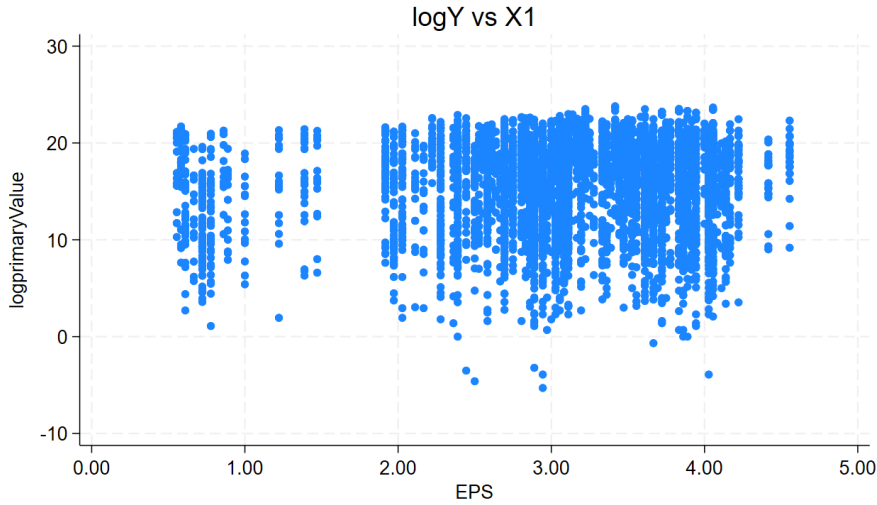


Figure 4: Relationship between log exports and EPS.

Figure 4 illustrates the relationship between export values and Environmental Policy Stringency (EPS). The visual evidence points to a weak and noisy association, with substantial heterogeneity across observations.

The expected sign of the relationship is theoretically ambiguous. On the one hand, stricter environmental regulation in high-income importing countries may increase demand for pollution-intensive imports if domestic production is constrained or relocated abroad. On the other hand, more stringent regulation may reduce overall consumption of pollution-intensive goods, leading to a negative association between exports and EPS.

2.2 GDP of exporting LMIC

Many studies include GDP as a control variable, and in some cases normalize trade flows by GDP. In this analysis, we include GDP as the focus of the study is closely related to the pollution haven hypothesis (Hu et al., 2011; Chakroun et al., 2020).

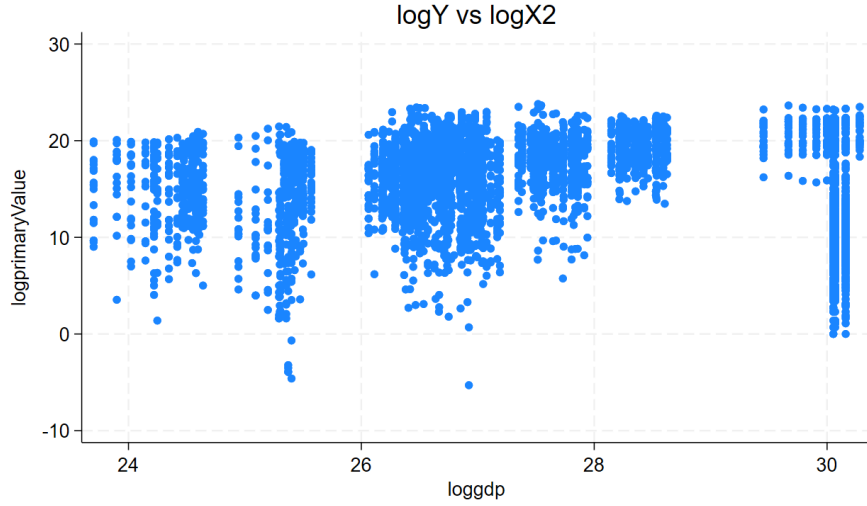


Figure 5: Relationship between log exports and log GDP.

Figure 5 illustrates the relationship between the logarithm of exports and the logarithm of GDP. While countries with larger economies tend to export higher values, the relationship appears highly heterogeneous across observations.

The expected sign of this relationship is positive, as larger low- and middle-income economies generally export greater volumes of goods overall, including pollution-intensive products.

2.3 Distance

The distance between 2 countries is seen as one of the key component of trade. It thus should be included in the model, as shown by Yotov and al. (2016).

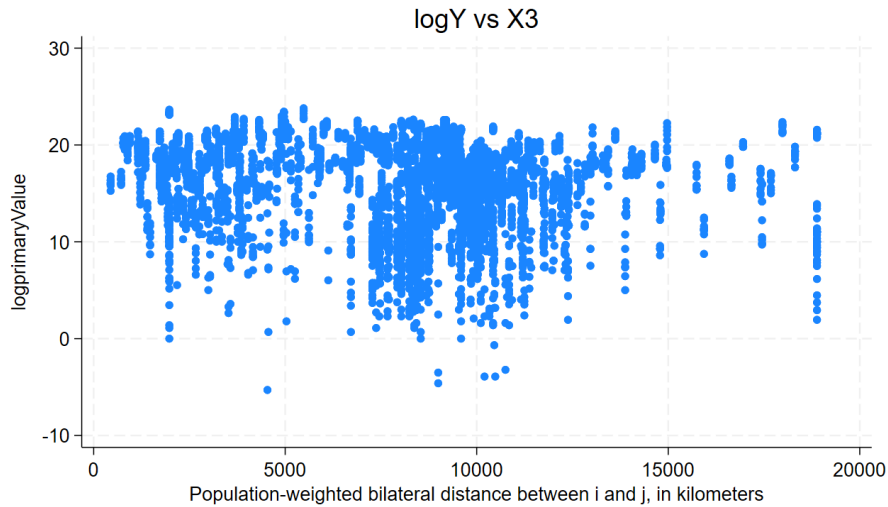


Figure 6: Relationship between log exports and distance.

Figure 6 depicts the relationship between export values and geographic distance. As expected, the relationship does not exhibit a clear linear pattern, and the observations are widely scattered.

The expected sign of the relationship is negative, as greater geographic distance is generally associated with lower bilateral trade flows due to higher trade costs.

2.4 Inflation in LMIC

Several articles show that inflation directly impacts the exchange rate and thus the exports of one country. Therefore, we use inflation rate data for each country (not both, to avoid having 2 variables correlated, and considering the structure of our database, it is easier to get inflation than exchange rate data) (Aliraqi, 2023; Çakmaklı, 2025). However, some articles show that inflation (in LMIC) negatively impacts their export, other (Yuliadi et al., 2024) say it has little impact (depending on the observed country group).

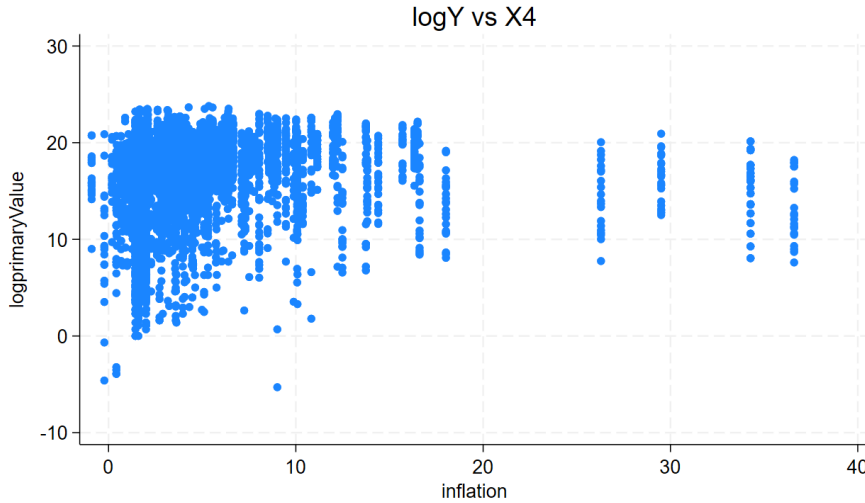


Figure 7: Relationship between log exports and inflation.

Figure 7 plots export values against inflation. The dispersion of observations indicates no clear direct relationship between inflation levels and export intensity.

The expected sign of this relationship is negative, as higher inflation tends to increase macroeconomic uncertainty and production costs, thereby reducing firms' ability to export.

3 Econometric Model & Identification Strategy

3.1 Model Specification

We estimate a log-linear panel data model of bilateral exports.

$$\log(Export_{ijt}) = \beta_1 EPS_{jt} + \beta_2 \log(GDP_{it}) + \beta_3 GeoDist_{ij} + \beta_4 CPI_{it} + \delta_j + \gamma_i + \omega_t + \epsilon_{ijt}$$

Where,

- EPS_{jt} is the environmental policy stringency in importing high-income country j ,
- $\log(GDP_{it})$ and CPI_{it} capture economic conditions in exporting LMIC i ,
- $GeoDist_{ij}$ captures bilateral trade costs,
- δ_j and γ_i are exporter and importer fixed effects,
- ω_t is a year fixed effect controlling for global shocks,
- ϵ_{ijt} is the error term.

We apply a logarithmic transformation to the dependent variable in order to interpret estimated coefficients as percentage changes rather than absolute unit changes. Within the gravity model framework, this implies a log-linear specification in which a one-unit increase in Environmental Policy Stringency (EPS) is associated with a $\beta_1 \times 100\%$ change in exports, *ceteris paribus*. This transformation mitigates the influence of large trade flows and ensures that the estimated effect of environmental regulation is proportional to the scale of trade, rather than being driven disproportionately by the largest exporting countries.

3.2 Estimation Method

We initially planned to estimate the model using exporter and importer fixed effects. This specification is motivated by the presence of unobserved, time-invariant country-specific characteristics—such as infrastructure quality, institutional performance, technological capabilities, or long-run environmental preferences—that may influence bilateral trade flows. In addition, year fixed effects are included to control for global shocks affecting all countries simultaneously, including commodity price fluctuations, macroeconomic cycles, and changes in international regulatory frameworks.

However, due to the multidimensional structure of the dataset and the large number of fixed effects to be accounted for simultaneously, the implementation of a standard panel

HDFE Linear regression	Number of obs	=	3,593
Absorbing 3 HDFE groups	F(4, 3539)	=	93.10
	Prob > F	=	0.0000
	R-squared	=	0.6849
	Adj R-squared	=	0.6802
	Within R-sq.	=	0.1248
	Root MSE	=	2.2231

logprimary~e	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
EPS	-.0788363	.1571286	-0.50	0.616	-.3869081	.2292354
loggdp	-.0695019	.3359727	-0.21	0.836	-.7282215	.5892177
DIST	-.0003458	.0000179	-19.28	0.000	-.000381	-.0003107
inflation	.004818	.013394	0.36	0.719	-.0214427	.0310788
_cons	21.62571	9.062331	2.39	0.017	3.857794	39.39363

Figure 8: HDFE Regression

fixed-effects estimator proved computationally inefficient. We therefore adopt a high-dimensional fixed-effects (HDFE) estimation approach, which allows exporter, importer, and year fixed effects to be absorbed in a computationally efficient manner without explicitly introducing a large set of dummy variables.

This estimator is particularly well suited for gravity-type trade models and ensures appropriate control for unobserved heterogeneity across countries and over time. Standard errors are computed using cluster-robust variance estimators.

4 Results

4.1 Baseline model

We begin our empirical analysis by estimating a baseline High-Dimensional Fixed Effects (HDFE) regression to test the immediate impact of environmental regulations on trade flows. This model accounts for time-invariant characteristics of both importers and exporters, as well as global shocks, through the inclusion of partner, reporter, and year fixed effects.

As shown in the table 8, the initial results do not provide statistical evidence for the Pollution Haven Hypothesis; the coefficient for Environmental Policy Stringency (EPS) is small and statistically insignificant ($p = 0.616$). DIST (Distance) is the most significant result ($P = 0.000$). The coefficient is -0.00034. Even though the number looks small, it confirms the gravity model theory: as distance increases, trade volume falls significantly. The coefficient on distance implies that an increase of 1 km in bilateral distance is associated with a 0.00034% decrease in exports. On the other hand, interestingly, GDP is also insignificant ($P = 0.836$). In many gravity models, exporter GDP is a huge driver of trade. The fact that it isn't significant here suggests that our Fixed Effects are absorbing most of the differences in economic size between these countries. Furthermore, we observe that the inclusion of the inflation variable significantly reduces our sample size—from 4,997 to 4,713 observations, due to missing data in key exporting countries

like Argentina. For the model fit and statistic part, we can see that R-squared is quite high, meaning that our model explains about 68.5% of the variation in exports.

4.2 Refined Analysis and Comparison

To better capture the dynamic relationship between environmental policy and trade, we refined our baseline model by making two key adjustments. First, we excluded the inflation variable; because of missing values and insignificance. By removing it, we regained significant data points, with price trends still being accounted by our year fixed effects. Second, we introduced one- and two-year lags for the EPS index. This acknowledges the reality that export contracts do not shift instantly, firms require time to react to new regulatory stringency in their destination markets.

The table 1 presents a comparison of these models side-by-side. Moving from the current-year baseline (Model 1) to the two-year lag (Model 3), we observe a clear trend: the negative impact of environmental regulations grows in both magnitude and statistical significance over time. In the two-year lag model, the coefficient for EPS reaches marginal significance at the 10% level ($p = 0.100$), suggesting that stricter regulations in high-income countries may eventually act as a barrier to entry for exporters in low- and middle-income countries rather than encouraging a relocation of dirty industries. Specifically, the coefficient of -0.261 in the two-year lag model implies that a one-unit increase in the environmental policy stringency of a high-income country is associated with an approximately 26.1% decrease in pollution-intensive exports from low- and middle-income countries two years later.

Regarding our control variables, the results for Distance and GDP are a mix of traditional theory and unexpected findings: For Distance, the coefficient remains negative and highly significant ($p = 0.000$) across all models. This is exactly what we expected to see in a gravity model: every additional kilometer significantly reduces the volume of pollution-intensive exports. On the other hand, the results for $\log(\text{GDP})$ are quite surprising. Theoretically, we expected a positive and significant sign, instead, coefficient is small and weakly significant, suggesting that the exporter's total economic size is not the primary driver of these trade flows. More precisely, the coefficient implies that a 1% increase in GDP is associated with a 0.778% decrease in exports

Table 1: Impact of Environmental Policy on Exports with Time Lags

	(1) Current	(2) 1-Year Lag	(3) 2-Year Lag
EPS	-0.112 (0.146)		
loggdp	-0.277 (0.316)	-0.509 (0.349)	-0.778* (0.400)
DIST	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
L.EPS		-0.146 (0.150)	
L2.EPS			-0.261* (0.158)
_cons	27.220*** (8.505)	33.765*** (9.401)	41.409*** (10.834)
N	3854	3347	2906

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 Limitations

The literature on environmental regulation and international trade considers carbon pricing as one of the main determinants of this topic, however, we could not include the CPS in our empirical model because of missing data. The CPS dataset, published by the OECD (8), is available only for three years - 2012, 2015, and 2018 - and does not have annual observations for the full period considered in the our study (2010-2019). On the other hand, because CPS is highly correlated with the EPS index, which we do include in our model, the exclusion of CPS does not really weaken our analysis. We can note that EPS captures a wide set of regulatory pressures (including carbon-related instruments) such that at least part of the information contained in CPS is still reflected in our main independent variable.

Another limitations would be introducing a two-year lag, this may still be too short to observe the full relocation of heavy industry. Building new factories in low-income countries is a long investment process. Our finding of a negative impact at $L2$ might represent the short-term shock of higher costs, while the actual relocation of production might only appear in a panel covering 15 or 20 years.

Finally, you might noticed that when you run the Stata do-file, the number of observations has changed. Actually It's due to missing data, it happen when we did logarithmic

transformations, and we did the panel construction. This data loss reflects standard empirical constraints rather than model misspecification.

6 Conclusion

Concluding this analysis, it has checked the Pollution Haven Hypothesis, taking into account whether stringent environmental policy in high-income countries drives exports of pollution-intensive goods from the low- and middle-income nations. Contrary to a prior expectation of "dirty" activities shifting to economies in developing countries, our empirical results found no support in this hypothesis within the immediate term.

Instead, our refined analysis uncovers a significant negative impact of regulations that is best described as emerging with a two-year lag. The more plausible interpretation is that stringent environmental measures act less like a "push factor" for relocation and more like a "Green Trade Barrier." Where high-income markets enhance their standards, it seems exporters in developing countries struggle either with compliance costs or with market access. The general conclusion that can be drawn from these findings should therefore be that environmental regulations are by no means limited to domestic policy instruments, but rather are becoming decisive factors for access to the globalized trade.

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